b) Develop a Java program to create an interface ParkingCharge containing the methods calculateCharge() and displayParkingDetails(). Create a class VehicleParking with members Vehicle Number, Vehicle Type, and Parking Hours. Derive the classes TwoWheeler and FourWheeler from VehicleParking and implement the interface. Calculate the parking charge as follows: For TwoWheeler, charge Rs. 20 per hour for the first two hours and Rs. 10 per hour for each additional hour. For FourWheeler, charge Rs. 50 per hour for the first two hours and Rs. 30 per hour for each additional hour. Display vehicle details and total parking charge. Demonstrate runtime polymorphism using a ParkingCharge reference.

```
interface ParkingCharge {
    double calculateCharge();
    void displayParkingDetails();
}

class VehicleParking {
    String vehicleNumber;
    String vehicleType;
    int parkingHours;

    public VehicleParking(String vehicleNumber, String vehicleType, int parkingHours) {
        this.vehicleNumber = vehicleNumber;
        this.vehicleType = vehicleType;
        this.parkingHours = parkingHours;
    }
}

class TwoWheeler extends VehicleParking implements ParkingCharge {

    public TwoWheeler(String vehicleNumber, int parkingHours) {
        super(vehicleNumber, "Two Wheeler", parkingHours);
    }

   public double calculateCharge() {
        if (parkingHours <= 2) {
            return parkingHours * 20.0;
        } else {
            return (2 * 20.0) + ((parkingHours - 2) * 10.0);
        }
    }

   public void displayParkingDetails() {
        System.out.println("Vehicle Number: " + vehicleNumber);
        System.out.println("Vehicle Type: " + vehicleType);
        System.out.println("Parking Hours: " + parkingHours + " hrs");
        System.out.println("Total Parking Charge: Rs. " + calculateCharge());
        System.out.println("1BM26CS483-T");
        System.out.println("CHANDANA BR");
        System.out.println("-----------------------------------");
    }
}
```

```java
class FourWheeler extends VehicleParking implements ParkingCharge {

    public FourWheeler(String vehicleNumber, int parkingHours) {
        super(vehicleNumber, "Four Wheeler", parkingHours);
    }

   public double calculateCharge() {
        if (parkingHours <= 2) {
            return parkingHours * 50.0;
        } else {
            return (2 * 50.0) + ((parkingHours - 2) * 30.0);
        }
    }

    public void displayParkingDetails() {
        System.out.println("Vehicle Number: " + vehicleNumber);
        System.out.println("Vehicle Type: " + vehicleType);
        System.out.println("Parking Hours: " + parkingHours + " hrs");
        System.out.println("Total Parking Charge: Rs. " + calculateCharge());
        System.out.println("-----------------------------------");
    }
}

public class ParkingManagementSystem {
    public static void main(String[] args) {

        ParkingCharge parkingRef;

        parkingRef = new TwoWheeler("KA-01-AB-1234", 5);
        parkingRef.displayParkingDetails();

        parkingRef = new FourWheeler("KA-05-CD-5678", 4);
        parkingRef.displayParkingDetails();
    }
}
```

```
----jGRASP exec: java ParkingManagementSystem
Vehicle Number: KA-01-AB-1234
Vehicle Type: Two Wheeler
Parking Hours: 5 hrs
Total Parking Charge: Rs. 70.0
1BM26CS483-T
CHANDANA BR
-----------------------------------
Vehicle Number: KA-05-CD-5678
Vehicle Type: Four Wheeler
Parking Hours: 4 hrs
Total Parking Charge: Rs. 160.0
1BM26CS483-T
CHANDANA BR
-----------------------------------

----jGRASP: Operation complete.
```